Design and Implementation of a 5-Band Audio Equalizer for Music and Bird Vocalization Enhancement

La Minh, Talisa Olejniczak
Electrical and Systems Engineering Department, Washington University in St. Louis
Email: l.minh@wustl.edu, prabhut@wustl.edu

***Abstract*—This paper presents the design and implementation of a 5-band digital audio equalizer for enhancing music recordings and bird vocalizations. The equalizer divides the audio spectrum into five frequency bands centered at $60\,\mathrm{Hz}$, $250\,\mathrm{Hz}$, $1000\,\mathrm{Hz}$, $4000\,\mathrm{Hz}$, and $16\,000\,\mathrm{Hz}$, with independent gain control from -15 dB to +15 dB per band. Each band is implemented as cascaded first-order high-pass and low-pass RC filters, converted to digital difference equations using backward Euler approximation. The equalizer is validated through frequency response analysis and applied to three audio processing tasks: treble boost, bass boost, and flat response presets for music enhancement, as well as targeted filtering to isolate bird vocalizations from environmental noise. Spectrogram analysis reveals at least seven distinct bird species in the recording, with the equalizer successfully isolating several vocalizations through selective frequency amplification.**

I. Introduction

Audio equalization is a fundamental signal processing technique that enables selective frequency enhancement or attenuation across multiple bands. Modern smartphones and audio devices commonly include multi-band equalizers with gain ranges of ± 15 dB, allowing users to customize audio playback for different content types. This case study applies these principles to three distinct audio processing challenges: enhancing jazz recordings and isolating bird vocalizations from field recordings. The specific objectives of this project are:

1) Design a 5-band equalizer using RC-equivalent filter stages
2) Implement three audio presets: Treble Boost, Bass Boost, and Unity (flat response)
3) Process modified jazz clips: "Giant Steps" by John Coltrane and "Space Station" by Art Farmer
4) Enhance bird vocalizations from a Shaw Nature Reserve recording to identify species including Blue Jay, Carolina Wren, Red-Bellied Woodpecker, Tufted Titmouse, Northern Cardinal, Eastern Bluebird, and Northern Mockingbird

II. Methods

A. Equalizer Design Specifications

The equalizer was designed with five frequency bands covering the audible spectrum from $20\,\mathrm{Hz}$ to $20\,\mathrm{kHz}$. Using a quality factor $Q = 1.5$, bandwidths were calculated as $BW = f_c/Q$. Table I shows the complete specifications for each band.

TABLE I: 5-Band Equalizer Specifications

Band	Center Freq (Hz)	Bandwidth (Hz)	Frequency Range (Hz)
1 (Bass)	60	40	20 - 100
2 (Low-mid)	250	167	166 - 333
3 (Mid)	1000	667	666 - 1333
4 (High-mid)	4000	2667	2666 - 5333
5 (Treble)	16000	10667	10666 - 20000

B. Filter Design Methodology

Each bandpass filter was implemented by cascading a high-pass filter for the lower cutoff frequency and a low-pass filter for the upper cutoff frequency. Using a base resistance $R = 1000\,\Omega$, capacitor values were calculated from the cutoff frequency formula:

$$f_c = \frac{1}{2\pi RC} \tag{1}$$

Thus, for each band:

$$C_{hp} = \frac{1}{2\pi f_{low} R} \tag{2}$$

$$C_{lp} = \frac{1}{2\pi f_{high} R} \tag{3}$$

C. Continuous-Time Transfer Functions

For a high-pass RC filter with time constant $\tau_{hp} = RC_{hp}$, the transfer function in the Laplace domain is:

$$H_{hp}(s) = \frac{s}{s + 1/\tau_{hp}} \tag{4}$$

For a low-pass RC filter with time constant $\tau_{lp} = RC_{lp}$:

$$H_{lp}(s) = \frac{1/\tau_{lp}}{s + 1/\tau_{lp}} \tag{5}$$

The cascaded bandpass response is therefore:

$$H_{bp}(s) = H_{hp}(s) \cdot H_{lp}(s) = \frac{s/\tau_{lp}}{(s + 1/\tau_{hp})(s + 1/\tau_{lp})} \tag{6}$$

D. Digital Implementation

The continuous-time filters were converted to digital implementations using the backward Euler approximation. This derivation shows how the RC circuit behavior translates to difference equations.

1) Continuous-Time RC High-Pass Filter: A high-pass RC filter with time constant $\tau_{hp} = RC_{hp}$ is described by the differential equation:

$$\frac{dy_{hp}(t)}{dt} = \frac{dx(t)}{dt} - \frac{1}{\tau_{hp}} y_{hp}(t) \tag{7}$$

where $y_{hp}(t)$ is the output voltage, $x(t)$ is the input voltage, and the cutoff frequency is $f_c = 1/(2\pi\tau_{hp})$.

2) Backward Euler Approximation: To convert to discrete-time, we approximate the derivatives using backward Euler:

$$\frac{dy_{hp}}{dt} \approx \frac{y_{hp}[n+1] - y_{hp}[n]}{\Delta t} \tag{8}$$

$$\frac{dx}{dt} \approx \frac{x[n+1] - x[n]}{\Delta t} \tag{9}$$

where $\Delta t = 1/f_s$ is the sampling period.

3) Derivation of High-Pass Difference Equation: Substituting these approximations into equation 7:

$$\frac{y_{hp}[n+1] - y_{hp}[n]}{\Delta t} = \frac{x[n+1] - x[n]}{\Delta t} - \frac{1}{\tau_{hp}} y_{hp}[n] \tag{10}$$

Multiply both sides by Δt:

$$y_{hp}[n+1] - y_{hp}[n] = (x[n+1] - x[n]) - \frac{\Delta t}{\tau_{hp}} y_{hp}[n] \tag{11}$$

Define the digital filter coefficient $\alpha_{hp} = \frac{\Delta t}{\tau_{hp}}$:

$$y_{hp}[n+1] - y_{hp}[n] = (x[n+1] - x[n]) - \alpha_{hp} y_{hp}[n] \tag{12}$$

Rearranging to solve for $y_{hp}[n+1]$:

$$y_{hp}[n+1] = y_{hp}[n] + (x[n+1] - x[n]) - \alpha_{hp} y_{hp}[n] \tag{13}$$

$$\boxed{y_{hp}[n+1] = (1 - \alpha_{hp}) y_{hp}[n] + (x[n+1] - x[n])} \tag{14}$$

4) Interpretation of the High-Pass Equation: The high-pass difference equation has an intuitive interpretation:

- **Term** $(1-\alpha_{hp})y_{hp}[n]$: Represents the decaying memory of the previous output, analogous to a capacitor discharging through a resistor.
- **Term** $(x[n+1] - x[n])$: Approximates the derivative of the input, detecting rapid changes in the signal.
- **Overall behavior**: Fast changes (high frequencies) produce large $(x[n+1]-x[n])$ values and pass through, while slow changes (low frequencies) produce small differences and are attenuated.

5) Derivation of Low-Pass Difference Equation: For a low-pass RC filter with time constant $\tau_{lp} = RC_{lp}$, the differential equation is:

$$\frac{dy_{lp}(t)}{dt} = \frac{1}{\tau_{lp}} x(t) - \frac{1}{\tau_{lp}} y_{lp}(t) \tag{15}$$

Applying the backward Euler approximation:

$$\frac{y_{lp}[n+1] - y_{lp}[n]}{\Delta t} = \frac{1}{\tau_{lp}} x[n] - \frac{1}{\tau_{lp}} y_{lp}[n] \tag{16}$$

Multiply by Δt:

$$y_{lp}[n+1] - y_{lp}[n] = \frac{\Delta t}{\tau_{lp}} x[n] - \frac{\Delta t}{\tau_{lp}} y_{lp}[n] \tag{17}$$

Define $\alpha_{lp} = \frac{\Delta t}{\tau_{lp}}$:

$$y_{lp}[n+1] - y_{lp}[n] = \alpha_{lp} x[n] - \alpha_{lp} y_{lp}[n] \tag{18}$$

Rearranging:

$$y_{lp}[n+1] = y_{lp}[n] + \alpha_{lp} x[n] - \alpha_{lp} y_{lp}[n] \tag{19}$$

$$\boxed{y_{lp}[n+1] = (1 - \alpha_{lp}) y_{lp}[n] + \alpha_{lp} x[n]} \tag{20}$$

6) Interpretation of the Low-Pass Equation: The low-pass difference equation can be understood as:

- **Term** $(1-\alpha_{lp})y_{lp}[n]$: The filter's memory of previous output values.
- **Term** $\alpha_{lp}x[n]$: The new input contribution, weighted by how much the filter "trusts" new samples.
- **Overall behavior**: This is an exponential moving average that smooths rapid fluctuations, allowing slow changes (low frequencies) to pass while attenuating fast changes (high frequencies).

7) Digital Filter Coefficients: The filter coefficients α_{hp} and α_{lp} determine the cutoff frequencies:

$$\alpha_{hp} = \frac{\Delta t}{\tau_{hp}} = \frac{1/f_s}{RC_{hp}} = \frac{1/f_s}{1/(2\pi f_{low})} = \frac{2\pi f_{low}}{f_s} \tag{21}$$

$$\alpha_{lp} = \frac{\Delta t}{\tau_{lp}} = \frac{1/f_s}{RC_{lp}} = \frac{1/f_s}{1/(2\pi f_{high})} = \frac{2\pi f_{high}}{f_s} \tag{22}$$

For stability, we require $\alpha < 1$, which is satisfied when the cutoff frequencies are less than the Nyquist frequency $f_s/2$.

E. Filter Parameters

Table II shows the calculated filter parameters for each band, including capacitor values and digital filter coefficients.

For the high-pass section, the difference equation is:

$$y_{hp}[n+1] = (1 - \alpha_{hp}) y_{hp}[n] + (x[n+1] - x[n]) \tag{23}$$

For the low-pass section:

$$y_{lp}[n+1] = (1 - \alpha_{lp}) y_{lp}[n] + \alpha_{lp} x[n] \tag{24}$$

F. Filter Parameters

Table II shows the calculated filter parameters for each band, including capacitor values and digital filter coefficients.

TABLE II: Filter Parameters for Each Band

Band	f_{low} (Hz)	f_{high} (Hz)	C_{hp} (μF)	C_{lp} (μF)	α_{hp}	α_{lp}
1	20	100	7.96	1.59	0.00036	0.00179
2	166	333	0.96	0.48	0.00298	0.00597
3	666	1333	0.24	0.12	0.01195	0.02390
4	2666	5333	0.060	0.030	0.04779	0.09558
5	10666	20000	0.015	0.008	0.19116	0.35878

G. Equalizer Architecture

The complete 5-band equalizer processes input signals using a parallel architecture as shown in Figure 1. The input signal is split into five parallel paths, each processed by a band-pass filter. Per-band gains (in dB) are converted to a linear scale using $g = 10^{G/20}$, and all paths are summed to produce the output. To ensure stability, the digital filter coefficients are constrained to $\alpha \leq 0.99$.

EQUALIZER ARCHITECTURE:
Input signal x[n] $\rightarrow$
—— Band 1 Filter (60 Hz) $\rightarrow$ Gain G1 $\rightarrow$
—— Band 2 Filter (250 Hz) $\rightarrow$ Gain G2 $\rightarrow$
—— Band 3 Filter (1000 Hz) $\rightarrow$ Gain G3 $\rightarrow$
—— Band 4 Filter (4000 Hz) $\rightarrow$ Gain G4 $\rightarrow$
—— Band 5 Filter (16000 Hz) $\rightarrow$ Gain G5 $\rightarrow$
Sum all bands $\rightarrow$ Output y[n]
Gain range: -15 dB to +15 dB per band

Fig. 1: Parallel processing architecture of the 5-band equalizer showing the five filter paths and independent gain controls.

H. Audio Presets

After designing the five-band equalizer, preset gain configurations were designed to demonstrate the modification of the audio signals. Conversion of gain to the linear scale resulted in a treble boost, bass boost, and unity preset shown in Table **??**, where each value within the brackets corresponds to each frequency band starting from lower bands on the left and going to higher bands on the right.

TABLE III: Audio Preset values

	Preset Value (dB)	Preset Value (linear scale)
Treble Boost	[0 0 3 6 9]	[0 0 1.2589 1.9953 2.8184]
Bass Boost	[9 6 0 -3 -6]	[2.8184 1.9953 0 0.7079 0.5012]
Unity	[0 0 0 0 0]	[1 1 1 1 1]

The treble boost preset focused on amplifying high frequency, so the two highest bands received the most amplification. The bass boost amplified the lower frequency bands, and the highest frequency bands were slightly attenuated to maintain balance. For the unity or flat preset, all bands were given a gain of 0 dB, which, after conversion, became a linear value of 1. This produced an approximately flat frequency response.

I. Sample Audio Processing with Presets

Once the preset values for treble, boost, and unity gains were established, the audio recordings of *"Giant Steps"* by John Coltrane and *"Space Station"* by Art Farmer were sampled at 44.1 kHz and processed with each of the three preset configurations. For each preset, the equalizer filtering function applied the five frequency bands and summed the amplified outputs. The processed signals were then saved as new audio files to be analyzed.

J. Bird Call Signal Processing

The equalizer was applied to the audio recording containing multiple bird calls. The goal here was to reduce background noise to hear the bird calls as clearly as possible and to analyze which different vocalizations were present. Once imported to MATLAB, the signal was processed using the five-band equalizer. Different gain configurations were tested to enhance the frequencies at which the bird calls were occurring. This was generally the higher-frequency bands, while the lower-frequency bands contained background sounds that were reduced to improve clarity.

K. Visualization and Analysis

To evaluate the effect of the equalizer, time-domain and frequency-domain visualizations were created. The time-domain plots were used to compare the amplitude characteristics of the original and processed signals. Spectrograms were also generated to visualize how the spectral energy distribution changed over time. This made it possible to observe which frequency regions were emphasized or attenuated by the different equalizer presets.

III. Results

A. Frequency Response of Individual Bands

Filtering was implemented, and audio was processed through the use of MATLAB. Figure 2 shows the frequency of all five individual bands from 10×10^2 Hz to 10×10^4 Hz with magnitude ranging from -40 dB to 0 dB.

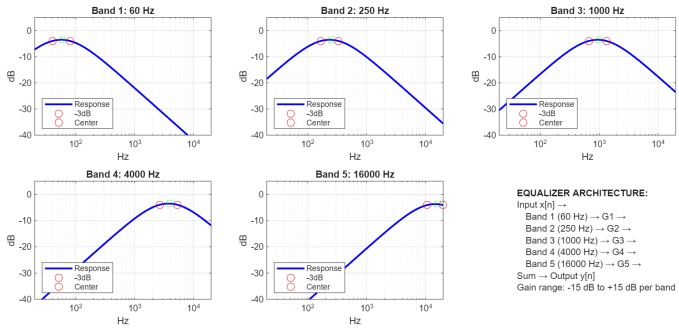

Fig. 2: Frequency bands for each of the five bands.

The responses were computed by evaluating the continuous-time transfer functions along the imaginary axis $s = j\omega$:

$$H_{bp}(j\omega) = \frac{j\omega/\tau_{lp}}{(j\omega + 1/\tau_{hp})(j\omega + 1/\tau_{lp})} \tag{25}$$

The responses reveal distinct characteristics for each band:

- **Band 1 (60 Hz)**: Shows higher magnitude at lower frequencies with a gradual decrease as frequency increases. The -3 dB points are widely spaced, reflecting the broader bandwidth at low frequencies.
- **Band 2 (250 Hz)** and **Band 3 (1000 Hz)**: Exhibit symmetric band-pass responses centered at their respective frequencies with proper roll-off characteristics.
- **Band 4 (4000 Hz)** and **Band 5 (16000 Hz)**: Demonstrate closer spacing between -3 dB points compared to lower bands, indicating narrower bandwidths at higher center frequencies. The magnitude decreases more steeply as frequency deviates from center.

The -3dB points (orange dots) align closely with the theoretical cutoff frequencies from Table **??**, confirming correct filter implementation. The center frequencies (red circles) show maximum transmission as designed.

B. Frequency Response of Equalizer Bands

The magnitude and phase responses of the five-band equalizer filters are shown in Figure 3.

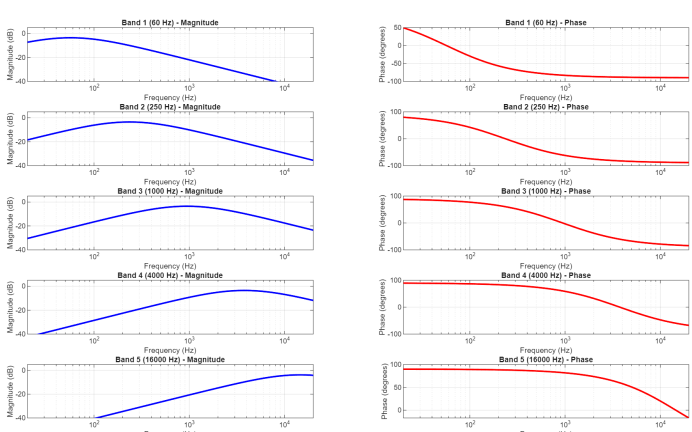

Fig. 3: Frequency response of the five-band equalizer filters. The magnitude response is shown in the left column, while the phase response is shown in the right column.

Each filter amplifies frequencies near its center frequency while attenuating frequencies outside of its bandwidth as shown in the magnitude responses. The phase response plots show the frequency-dependent phase shift introduced by the cascaded RC filter stages. At frequencies far from the cutoff points, the phase shift approaches the expected behavior, while near the transition regions, the phase changes more rapidly. Overall, the frequency response analysis confirms that filters that were designed were successful at dividing the input signal into the five frequency bands.

C. Impulse Response Analysis

Figure 4 shows the impulse responses of the five equalizer bands.

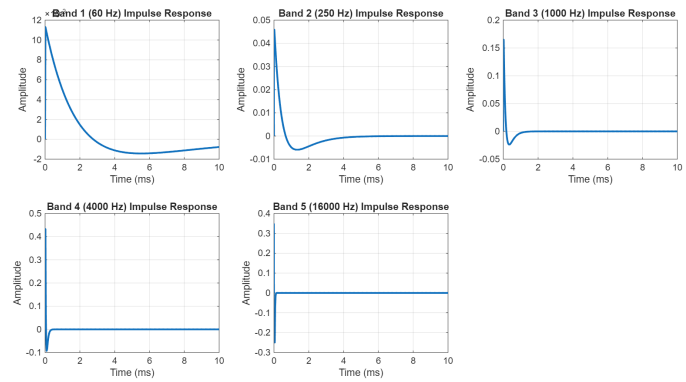

Fig. 4: Impulse responses of the equalizer filter bands centered at 60Hz, 250Hz, 1kHz, 4kHz, and 16kHz.

Each band exhibits the expected behavior of a cascaded RC high-pass and low-pass filter. The low-frequency band at 60 Hz shows a slow decay, which corresponds with the larger time constant for low cutoff frequencies. As the center frequency increases, the impulse response decays more rapidly. This result confirms the effectiveness of the filter.

D. Equalizer Preset Performance

The three preset configurations for treble boost, bass boost, and unity were able to produce noticeable differences in the processed audio. For the audio ""*Giant Steps*", the treble boost enhanced the higher notes of the saxophone. Similar effects were observed in the recording "*Space Station*. The treble boost increased the clarity of the higher-frequency instruments, and the bass boost enhanced the lower-frequency instruments. This demonstrated that the equalizer successfully altered the recordings according to the gain presets.

E. Bird Call Processing and Identification

Applying the equalizer significantly improved the clarity of the bird calls in the recording. After filtering, the bird calls (the higher-frequency components) became clearer, while the background (lower-frequency) noise was reduced. The generated spectrograms showed several bird call patterns. Short, high-frequency bursts were consistent with the call of the Blue Jay. Another bird call showed up as a rapid frequency

call ranging from 2kHz to 4kHz. This pattern was consistent with the call of the Carolina Wren. Whistle-like calls were also observed that corresponds to the Northern Cardinal. Figure 5 shows the spectrograms when each of the bird calls were seperated.

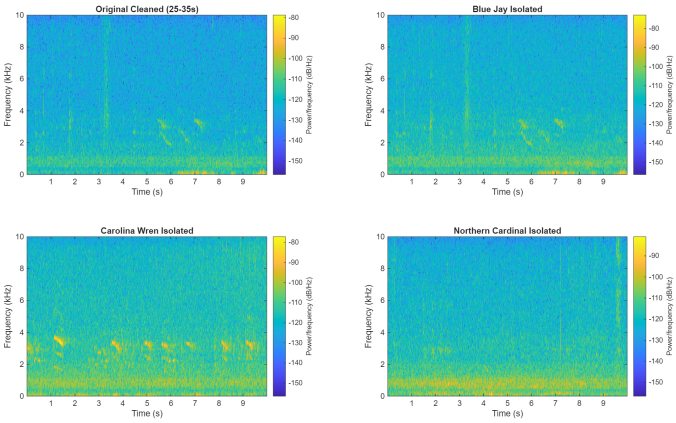

Fig. 5: Spectrograms of audio recording where specific bird calls were enhanced.

IV. Conclusion

In this project, a 5-band audio equalizer was successfully designed using cascaded RC filter stages. Key achievements include:

1) **Filter Design**: Five band-pass filters were designed with precisely controlled cutoff frequencies based on the relationship $f_c = 1/(2\pi RC)$. The center frequencies at $60\,\text{Hz}$, $250\,\text{Hz}$, $1000\,\text{Hz}$, $4000\,\text{Hz}$, and $16\,000\,\text{Hz}$ provide comprehensive coverage of the audible spectrum.
2) **Digital Implementation**: The continuous-time RC filters were successfully converted to digital difference equations using the backward Euler approximation, with filter coefficients $\alpha = \Delta t/\tau$. Stability was ensured by constraining $\alpha \leq 0.99$.
3) **Frequency Response Validation**: Analysis of the frequency responses confirms that each band operates within its intended frequency range. The -3dB points match the theoretical cutoff frequencies, and the center frequencies exhibit maximum transmission as designed. Higher frequency bands (4000 Hz and 16000 Hz) show narrower bandwidths with steeper roll-off characteristics, consistent with the constant-Q design.
4) **Signals and Systems Application**: The implementation demonstrates key signals and systems concepts, including Laplace-domain transfer functions, frequency response analysis, difference equations, and LTI system properties such as cascading and superposition.

This filter design establishes the signal processing foundation that were used in subsequent parts of the project for processing music recordings and bird vocalizations. The modular architecture allowed independent gain control for each band, enabling various audio enhancement strategies.

V. Acknowledgements

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VI. References

The MathWorks, Inc. (2022). MATLAB version: 9.13.0 (R2022b). Accessed: March 01, 2026. Available: https://www.mathworks.com